

WATERLINE

The Global Waterfront-Only Property Portal

Complete Technical Specification, Build Brief & Operating Manual — Version 2.1 (definitive)

Working codename: WATERLINE. Alternatives held in reserve: TIDELINE, SHOREBOUND, RIVAGE, WATERMARK ESTATES. Replace the codename in one place only — see §9.1.

0. How to read this document

This is the complete brief for a new property portal listing **only properties with direct water access, worldwide**. It is written for one senior full-stack developer working with **Cursor + Claude**, and it is designed so that the developer can reach **90-100 % of a launch-ready site without asking the client a single question**.

Everything a developer would normally have to ask has been pre-answered:

Normally asked	Answered in
"What copy goes here?"	§11 Copy deck — complete English microcopy
"What colours, fonts, spacing?"	§9 Design tokens — exact values
"How should agencies upload listings?"	§8 Multi-agency backend — full workflow, CSV template, feed contract
"Where do I get photos, copy, geodata, market figures?"	§13 Content sourcing — every asset, named source, licence position
"Which decisions are mine to make?"	§21 Pre-answered decisions — 45 defaults, all reversible
"When do I stop and ask?"	§1.5 Escalation rules — only 6 things require the client

What changed in v2.x

v2.0 added: the developer operating method with Claude/Cursor (§1), a full **multi-agency backoffice** with roles, approval workflow, bulk import and feed ingestion (§8), a **complete copy deck** (§11), **AI/answer-engine optimisation** (§14.6), **accessibility to WCAG 2.2 AA** (§15), anti-scraping and data retention (§16), runbooks (§19), **45 pre-answered product decisions** (§21), and expanded Cursor prompts with explicit acceptance criteria (§22). The data model gained the multi-tenancy, moderation, duplicate-detection and listing-lifecycle fields that v1.0 lacked.

v2.1 adds **§13 Content sourcing and production** — a complete inventory of every asset the site needs (photography, geodata, market data, fonts, logo, editorial, translations, legal text) with a named source, a licence position and a production method for each, plus the two-track delivery plan in §2.5 that separates the fast public launch from the multi-agency backoffice.

Reading order: §1 → §2 → §6 → §8 → §12 → §21 → §22. Everything else is reference while building.

1. How the developer builds this (the operating method)

The target is 90-100 % complete before the client is involved. That requires discipline about *how* Cursor and Claude are used, not just what is pasted into them.

1.1 Repository setup for AI-assisted development

Create these three files in the repo **before writing any feature code**. They are what keeps a long AI-assisted build coherent.

`/CLAUDE.md` — the project constitution, read by Claude Code and Cursor on every task:

```
# WATERLINE — project rules for AI assistants

## What this is
A worldwide property portal that lists ONLY properties with direct water access
(sea, ocean, lake, river, lagoon, canal, fjord). Next.js 15 App Router + Payload 3
+ Postgres/PostGIS + Typesense, deployed on Vercel.

## Non-negotiable rules
1. Performance budgets are acceptance criteria, not goals. Home first-load JS < 110KB
  gzip, listing < 130KB, search < 160KB. LCP < 1.2s mobile. Never add a dependency
  that pushes a route over budget — check with `pnpm analyze` before committing.
2. No component or route queries the database directly. Everything goes through
  /src/lib/db with explicit return types.
3. No hardcoded colours, spacing, font sizes or radii outside /src/tokens. ESLint
  enforces this; do not disable the rule.
4. Every public page is SSG or ISR. `force-dynamic` is forbidden on indexable routes.
5. Controlled enums over free text, always. If a field could be an enum, it is one.
6. The water rule: a listing cannot publish without >=1 waterAccessType and
  distanceToWaterM <= 50. Never weaken this validation.
7. All user-facing strings come from /src/messages/*.json. No literal strings in JSX.
8. Sample/demo listings always carry isSample=true, render a SAMPLE badge, emit
  noindex, and are excluded from sitemaps.
9. Accessibility: WCAG 2.2 AA. Every interactive element keyboard-reachable with a
  visible focus ring. Never ship a div with onClick.
10. Multi-agency: every query that returns listings must respect agency scoping and
  moderation status. An `agent` role never sees another agency's data.

## Definition of done for any task
- TypeScript compiles with zero errors (`pnpm typecheck`)
- ESLint passes (`pnpm lint`)
- Relevant tests pass (`pnpm test`), and new logic has a test
- `pnpm analyze` shows no route over its JS budget
- The feature works with JavaScript-heavy throttling (Slow 4G, 4x CPU)
- No console errors or hydration warnings

## Commands
pnpm dev · pnpm build · pnpm typecheck · pnpm lint · pnpm test · pnpm test:e2e
pnpm analyze · pnpm payload generate:types · pnpm seed · pnpm sample:purge
pnpm search:reindex · pnpm audit:seo · pnpm audit:all

## Conventions
- Server Components by default; 'use client' only where interaction requires it.
- Data fetching in Server Components or route handlers, never in useEffect.
- Zod schemas shared client/server, colocated in /src/lib/schemas.
- File naming: kebab-case files, PascalCase components, camelCase functions.
- Commits: conventional commits (feat:, fix:, perf:, chore:, docs:).
```

`/cursor/rules/waterline.mdc` — same rules, plus: "When asked to implement a section of the spec, first restate the acceptance criteria from the spec, then implement, then self-check against them and report pass/fail per criterion."

`/DECISIONS.md` — an append-only log. Every time the developer makes a judgement call not covered by §21, they add: date, decision, alternatives considered, reason, reversibility. This is what the client reviews at handover instead of being interrupted 45 times.

1.2 The build loop (repeat per prompt in §22)

1. **Read** the prompt and the spec sections it references.
2. **Plan** — ask Claude to produce an implementation plan and a file list *before* writing code. Reject the plan if it touches files outside its scope.
3. **Implement** in a feature branch (`feat/07-search-page`).
4. **Self-verify** against the prompt's acceptance criteria — the prompt contains them explicitly; make Claude report pass/fail per line.
5. **Measure** — `pnpm typecheck && pnpm lint && pnpm test && pnpm analyze`. For UI work, run Lighthouse on the affected route.
6. **Commit and tag** (`v0.7-search`). Small, reversible commits beat one giant diff.
7. **Log** any decision in `DECISIONS.md`.

1.3 Context management

Keep open in Cursor: `CLAUDE.md`, the relevant spec section, `/src/tokens/tokens.ts`, `/src/payload/collections/Property.ts`, and the file being edited. Do **not** paste the whole specification into every prompt — it dilutes attention. Reference sections by number; the prompts in §22 are self-contained.

When the context window degrades (repeated mistakes, forgotten conventions), start a new chat and re-anchor with: "Read `CLAUDE.md` and `/src/tokens/tokens.ts`. We are implementing §X of the spec. Here is the current file."

1.4 Quality gates that run without the client

- **CI on every PR:** typecheck, lint, unit tests, Playwright critical paths, Lighthouse CI budgets, bundle-size assertions, `audit:seo` (canonicals, hreflang, JSON-LD validity, sitemap coverage), `audit:all` (axe-core on 8 key routes).
- **Weekly self-review:** run `pnpm audit:all` and fix everything before continuing to new features. Debt compounds faster in AI-assisted builds than in hand-written ones.

- **Visual regression:** Playwright screenshots of 10 key viewports committed as baselines; any diff must be intentional.

1.5 The only six things that require the client

Everything else is decided by §21 or by the developer, logged in `DECISIONS.md`.

1. Final brand name, domain and logo approval.
2. Legal entity details for the footer, privacy policy and terms (company name, address, VAT, jurisdiction).
3. Go-live date and DNS access.
4. Payment/commercial model if listings are ever to be paid (Phase 1 assumes free).
5. Real agency and agent contact details for the first live listings.
6. Sign-off on the staging site before DNS cutover.

2. Product definition

2.1 One-line definition

A global search engine for property you can step off and into the water — sea, lake, river, lagoon, canal or fjord — with the world's only taxonomy built for people who own boats.

2.2 The hard rule (enforced in code)

A listing is admissible **only** with at least one qualifying `waterAccessType` **and** `distanceToWaterM ≤ 50`. "Sea view", "500 m from the beach", "walking distance to the lake" are rejections. Enforced by a Payload validation hook, surfaced to the submitting agency with a clear reason, and logged.

This rule is the brand. If it leaks, we become a worse JamesEdition.

2.3 Audiences, ranked

Audience	What they search for	Feature that serves them
Boat & yacht owners	"Will my 24 m boat fit, and can I reach open sea?"	Berth / draft / bridge-clearance filters — nobody else has this
Trophy-asset buyers	Beauty, privacy, exclusivity	Editorial design, full-bleed imagery, discretion controls
Lifestyle relocators	Swimmable? Sandy? Sunset-facing?	Beach type, orientation, water-quality fields
Investors & developers	Frontage metres, buildability, tenure	Linear frontage, shoreline tenure, concession fields
Agencies & brokers (supply side)	Free reach to qualified international buyers	Self-service backoffice, bulk upload, XML feeds (§8)

The supply side is a first-class audience in v2.0. A listings portal with no inventory is worthless, and inventory arrives only if uploading is effortless.

2.4 Scope of Phase 1

In: public site (home, map search, listing detail, programmatic landing pages, destination hub, journal, editorial, contact), **multi-agency backoffice with approval workflow, bulk import and feed ingestion**, sample-listing generator, 6 languages, full SEO + AI-discovery layer, accessibility to AA, analytics, lead capture and routing, performance pass, staging and launch.

Out (seams left in place): end-user accounts and saved searches, alerts, paid listings/billing, CRM integration, AI semantic search, native apps, agent messaging inbox.

2.5 Two delivery tracks — do not mix them

Track A — public launch (Prompts 1-10, 12-20 of §22). Everything a visitor sees, plus the sample inventory, SEO, performance, accessibility and deploy. This is the track that gets a complete, impressive, indexable site live fast, and it depends on nothing outside the repository.

Track B — multi-agency supply (Prompt 11 of §22, plus §8). Roles, onboarding, bulk import, feeds, moderation queue, expiry sweep, agency dashboard. Every field it needs already exists in the data model from Prompt 3, so Track B can be built **before or after go-live without a migration or a rewrite**.

Build Track A end to end first. If the client wants inventory from external agencies on day one, run Track B before launch; if not, ship Track A and add Track B in the following sprint. The spec is written so that either order works and neither delays the other.

3. Benchmarks — what we take and from whom

The developer should open each of these on a real phone before writing code.

3.0 Legal boundary

We replicate **interaction patterns and information architecture** — conventions, not property. We never copy HTML/CSS/JS, scraped markup, photography, logos, licensed fonts, listing copy, editorial text, or a competitor's listing database. Scraping a competitor's inventory is both a terms-of-service breach and a legal risk, and it is unnecessary (\$13). "As good as JamesEdition" is the target; "looks like JamesEdition" is not.

3.1 Primary references

JamesEdition — https://www.jamesedition.com/real_estate · waterfront vertical https://www.jamesedition.com/real_estate/waterfront--all The scale benchmark: 200,000+ listings, 16,000 luxury offices, 50+ countries, and vertical category pages (beachfront villas, ski chalets, vineyard estates, private islands) that function as a lifestyle-facet taxonomy. **Take:** category-as-landing-page model; dense but calm grid; price and location as the primary card label; the editorial "stories" layer feeding organic traffic. **Note:** their waterfront vertical is our entire site — and you cannot filter it by frontage, berth or water body. That gap is our product. **Avoid:** heavy third-party script load and marketplace-generic listing pages.

Christie's International Real Estate — <https://www.christiesrealestate.com/lifestyles/waterfront-homes/> The reference for editorial restraint: generous whitespace, serif display type, muted palette, and a clean [lifestyles/](#) URL structure. **Avoid:** slow hero video and a long scroll before the first listing.

Sotheby's International Realty — <https://www.sothebysrealty.com/> · **The Waterfront Property** — <https://www.thewaterfrontproperty.com/> The second is a Sotheby's-network waterfront specialist presenting curated beachfront, seafront, sea-view and lake property across 83 countries — the closest positioning to ours. **Take:** curated-collection framing, country navigation as first-class, agent trust signals. **Avoid:** their blurring of "waterfront" with "sea view" — the exact dilution we reject.

Private Islands Online — <https://www.privateislandsonline.com/> · **Vladi Private Islands** — <https://www.vladi-private-islands.de/en/> Hyper-niche water portals with real authority (Vladi: 2,400+ certified island sales) and real faceted search (availability, price range, development state, island lifestyle facets). **Take:** niche-authority tone, island facet vocabulary, the sold archive as trust + SEO asset. **Avoid:** dated UI, weak mobile, slow images. The market gap is exactly this: niche credibility with 2010-era execution.

Buy Waterfront Worldwide — <https://www.buywaterfrontworldwide.com/> The clearest statement of our positioning — a destination dedicated exclusively to waterfront property globally, organised by named destinations (French Riviera, Amalfi Coast, Santorini, Mallorca, Turks & Caicos, the Hamptons). **Take:** their destination list is a ready-made first batch of landing pages. Their thin content and absent search are the opportunity.

3.2 Pattern references (mechanics, not aesthetics)

Site	URL	Take exactly this
Airbnb	airbnb.com	Map + list split, "search as I move the map", sticky filter pills, in-card carousel
Zillow	zillow.com	Draw-your-own-area search, instant facet counts, URL-encoded filter state
Rightmove	rightmove.co.uk	Information density and speed on mobile; keyword-within-results
Idealista	idealista.com	Multi-locale architecture and per-locale URL hygiene
Compass	compass.com	Listing-page rhythm: gallery → facts → story → map → agent
YachtWorld / Boat International	yachtworld.com · boatinternational.com	The vocabulary of boat owners: LOA, beam, draft, berth — source of our differentiating filters
Linear / Vercel / Stripe	linear.app · vercel.com · stripe.com	Perceived performance: instant transitions, skeletons, zero layout shift
Notion / Framer (marketing pages)	notion.com · framer.com	Editorial section rhythm and type scale on long pages

3.3 The competitive thesis

Incumbents are either **broad and polished** (JamesEdition, Christie's) or **niche and dated** (Vladi, Private Islands, Buy Waterfront Worldwide). Nobody is niche, polished, fast and structured around water data. The moat is the data model (\$6): expensive for others to retrofit, free for us to own from day one.

4. Design and UX decisions (settled — do not re-open)

1. **Editorial luxury, not portal chrome.** Christie's restraint plus JamesEdition density. Large type, deep whitespace, no gradients, no card drop-shadows, no pill-everything. Photography carries emotion; the UI gets out of the way.

2. **Water is the visual system.** Deep teal/ink palette with a warm sand accent (§9.2). A recurring 1 px horizon rule at section breaks.
3. **The map is a surface, not a tab.** Persistent split view on desktop, toggle on mobile. Custom style: water saturated, land desaturated.
4. **Cards show water data.** Price · location · **frontage (m)** · water body · access-type icon. Beds and baths are secondary. This alone signals specialism in three seconds.
5. **"Fits my boat" is the hero filter.** A boat-length input sits on the homepage beside price and location. Most screenshot-able feature on the site.
6. **No hero carousel.** One still image, or a muted ≤ 1.5 MB AV1 loop with a static poster. LCP depends on it.
7. **No cookie wall before LCP.** Consent is non-blocking, deferred, never layout-shifting.
8. **Progressive disclosure on detail pages,** with the water-credentials block as a designed table, not bullets.
9. **Discretion controls exist on every listing** (approximate coordinates, hidden address, unlisted visibility) — luxury sellers require them, and they are three fields plus one map behaviour.
0. **Empty states are computed, never generic:** always name the most restrictive filter and offer one click to relax it.

5. Information architecture and URL scheme

5.1 Locales

Phase 1 ships **6 locales**: `en` (default), `it`, `fr`, `de`, `es`, `ru`. Adding a locale must require only a config entry and a message file. Locale prefix is always present; `/` performs a 302 language-detect redirect and is never cached.

5.2 Public routes

<code>/[locale]</code>	Home
<code>/[locale]/search</code>	Map + list search (state in query string)
<code>/[locale]/property/[slug]</code>	Listing detail
<code>/[locale]/waterfront/[combo]</code>	Programmatic landing page (§5.3)
<code>/[locale]/destinations</code>	Destination hub
<code>/[locale]/destinations/[slug]</code>	Destination page
<code>/[locale]/collections/[slug]</code>	Curated editorial collections
<code>/[locale]/journal · /journal/[slug]</code>	Editorial content
<code>/[locale]/agencies/[slug]</code>	Agency profile (SEO + supply-side trust)
<code>/[locale]/list-with-us</code>	Supply-side landing + application form
<code>/[locale]/about · /contact</code>	Editorial
<code>/[locale]/legal/[slug]</code>	Privacy, cookies, terms, disclaimer
<code>/[locale]/404 · /500</code>	Designed error states

5.3 Backoffice and API routes

<code>/admin/*</code>	Payload admin (agencies, agents, editors, admins)
<code>/api/health · /api/revalidate</code>	Ops
<code>/api/og/[type]/[id]</code>	Dynamic OG images
<code>/api/property/[slug]/brochure.pdf</code>	Brochure
<code>/api/feeds/[agencySlug]</code>	Inbound feed endpoint (token-authenticated)
<code>/api/leads</code>	Lead intake (rate-limited)
<code>/sitemap.xml + /sitemaps/*.xml</code>	Sitemap index and children
<code>/robots.txt · /llms.txt</code>	Crawler and AI-discovery policy (§14.6)

5.4 Programmatic landing-page grammar

Canonical: `/[locale]/waterfront/[propertyType]-[waterAccess]-[location]`, any segment omissible.

```
/en/waterfront/villas-sea-liguria
/en/waterfront/private-dock-homes-florida-keys
/en/waterfront/lakefront-villas-como
/en/waterfront/private-islands-caribbean
/en/waterfront/liguria (location only)
/en/waterfront/private-islands (type only)
```

Rules: exactly one canonical URL per combination; a resolver in `/lib/seo/combos.ts` normalises aliases and 301s; slug segments come from the controlled `Taxonomy` collection, localised per locale; a combination renders only when a `LandingPage` record exists with `published = true` **and** non-empty editorial `intro` — otherwise 404. Shipping empty combination pages is the fastest route to a Helpful-Content demotion.

5.5 Search URL contract

```
/en/search?water=sea,lake&access=private_dock&minFrontage=25&boatLoa=24&draft=2.5
&price=2000000-10000000&type=villa,estate&beds=4&bbox=8.1,44.2,9.6,44.6
&sort=frontage_desc&page=2
```

Paginated search pages are `noindex, follow` beyond page 1; the canonical points to page 1 of the same filter set. Only landing pages are indexable representations of filtered inventory — this prevents infinite crawl surfaces.

6. Data model

Defined once in Payload 3; TypeScript types generated from it are the single source of truth for API, DB and frontend. Postgres 16 with **PostGIS enabled from day one**.

6.1 `Property` — identity, ownership, lifecycle

Field	Type	Notes
<code>id</code>	uuid	
<code>slug</code>	text unique indexed	SEO-critical, immutable after first publish; changes create a <code>Redirect</code>
<code>reference</code>	text	Agency's own code; unique per agency
<code>agency</code>	relation → <code>Agency</code> , required, indexed	Multi-tenancy key. Every query is scoped by it.
<code>agent</code>	relation → <code>Agent</code> , nullable	Lead routing target
<code>status</code>	enum	<code>draft</code> · <code>pending_review</code> · <code>in_market</code> · <code>under_offer</code> · <code>sold</code> · <code>withdrawn</code> · <code>expired</code> · <code>archived</code>
<code>moderation</code>	enum	<code>unreviewed</code> · <code>approved</code> · <code>rejected</code> · <code>changes_requested</code>
<code>moderationNote</code>	text	Shown to the agency, not public
<code>visibility</code>	enum	<code>public</code> · <code>unlisted</code> · <code>private</code> (unlisted = link-only, noindex, out of sitemap)
<code>isSample</code>	bool default false	Demo data flag (§13.4)
<code>featured</code>	bool	Editorial only; agencies cannot self-feature
<code>publishedAt</code> / <code>expiresAt</code>	timestamp	Listings expire after 180 days without agency confirmation (§8.7)
<code>lastVerifiedAt</code>	timestamp	Set when an agency confirms the listing is still available
<code>duplicateOf</code>	relation → <code>Property</code> , nullable	Set by duplicate detection (§8.8)
<code>fingerprint</code>	text indexed	Deterministic hash for duplicate detection
<code>sourceType</code>	enum	<code>manual</code> · <code>csv_import</code> · <code>xml_feed</code> · <code>api</code>
<code>viewCount</code> / <code>leadCount</code>	int default 0	

6.2 Commercial

`priceType` enum `fixed` · `on_request` · `auction` · `poa` · `priceAmount` numeric nullable · `currency` enum `EUR` · `USD` · `GBP` · `CHF` · `AED` · `SGD` · `priceEur` numeric **generated on save from a daily FX snapshot**, indexed (the only field ever used for sorting and range filters) · `priceQualifier` enum `guide` · `asking` · `offers_over` · `reduced` · `tenure` enum `freehold` · `leasehold` · `concession` · `fractional` · `share_transfer` · `leaseYearsRemaining` int nullable · `serviceChargeAnnual` numeric nullable · `propertyTaxAnnual` numeric nullable · `availableFrom` date nullable.

6.3 Physical

`propertyType` enum: `villa` · `apartment` · `penthouse` · `estate` · `farmhouse` · `chalet` · `townhouse` · `castle` · `lighthouse` · `boathouse` · `private_island` · `land_plot` · `marina_residence` · `development_project` · `hotel_resort`.

`bedrooms` · `bathrooms` · `builtAreaSqm` · `plotAreaSqm` · `terraceAreaSqm` · `yearBuilt` · `renovatedYear` · `floors` · `parkingSpaces` · `condition` enum `new` · `renovated` · `good` · `to_renovate` · `shell` · `energyRating` text nullable · `features` multi-enum (`pool` · `infinity_pool` · `heated_pool` · `gym` · `spa` · `sauna` · `staff_quarters` · `helipad` · `tennis` · `vineyard` · `olive_grove` · `solar` · `geothermal` · `elevator` · `gated` · `smart_home` · `guest_house` · `garage` · `wine_cellar` · `cinema`).

6.4 Water credentials — the differentiator (mandatory block)

Field	Type	Values
waterBodyType	enum required indexed	sea · ocean · lake · river · lagoon · canal · fjord · bay · estuary · reservoir · marina_basin
waterBody	relation → WaterBody	Controlled: "Ligurian Sea", "Lake Como", "Intracoastal Waterway"
waterAccessType	multi-enum required min 1 indexed	private_beach · shared_beach · direct_shore · private_dock · private_mooring · marina_berth_included · boathouse · slipway · seawall_quay · rock_platform · riparian_access · whole_island
distanceToWaterM	int required 0-50	>50 blocks publication
waterFrontageM	numeric indexed	Linear private shoreline — headline card stat
beachType	enum	sand · pebble · rock · mixed · none
orientation	enum	N · NE · E · SE · S · SW · W · NW → "sunset-facing" filter
swimmableFromProperty	bool	
tidal / tideRangeM	bool / numeric	
waterfrontProtection	enum	natural · seawall · riprap · breakwater · pier_only
floodZone	text	Local designation verbatim + disclaimer
shorelineTenure	enum	private_to_waterline · private_to_high_water · public_easement · state_concession · riparian_rights — in Italy this is the <i>demanio marittimo</i> question and it materially changes value
concessionExpiry	date	Only when state_concession

6.5 Nautical block — the "fits my boat" engine

mooringType enum none · buoy · jetty · pontoon · fixed_dock · floating_dock · boat_lift · dry_dock · marina_berth · berthCount int · maxBoatLoaM numeric **indexed** · maxBoatBeamM numeric · waterDepthAtBerthM numeric **indexed** · navigableToOpenSea bool indexed · fixedBridgesToOpenSea bool · minBridgeClearanceM numeric · nearestMarinaName text · nearestMarinaDistanceKm numeric · shorePower bool · freshWaterAtDock bool · fuelDockNearby bool · helipad bool · seaplaneAccess bool.

These fifteen fields are worth more than any AI feature. They create a search no competitor can answer, and they are why a broker with a €30 M dock property sends us the listing first.

6.6 Location

location group: label · addressLine (admin-only, never public) · locality · province · region · country (ISO-3166-1 alpha-2, indexed) · continent · coordinates **PostGIS geography(Point,4326) required, GIST-indexed** · coordinatePrecision enum exact · approximate_500m · hidden · destination relation → Destination.

Privacy rule: when precision is approximate_500m, the public API returns a deterministically jittered point (seeded by listing id, so it never moves between loads) and the map draws a circle, never a pin. Exact coordinates must never reach the client for those listings — verify in the network tab, not just the UI.

6.7 Content (localised)

title · subtitle · description rich text · highlights (≤5 short strings) · media ordered relation → Media with localised alt · videoUrl · virtualTourUrl · floorplans · documents (private, admin/agency only) · metaTitle · metaDescription with auto-fallback (\$14.2).

6.8 Supporting collections

- Media** — variants generated at upload (320/640/960/1280/1920/2560, AVIF + WebP), EXIF stripped, blurhash LQIP, credit / licence / sourceUrl, width/height, dominantColor, plus agency scoping. Minimum accepted: 1600 px on the long edge; below that the upload is rejected with a clear message.
- Agency** — name, slug, logo, description (localised), country, languages, website, phone, WhatsApp, email, licence number, verified bool, verifiedAt, tier enum (standard · verified · partner), listingQuota int, feedUrl, feedToken, feedFormat enum, feedLastRunAt, feedLastStatus, billing fields reserved and empty.
- Agent** — name, photo, role, languages, phone, WhatsApp, email, agency relation, receivesLeads bool.
- Lead** — name, email, phone, message, property, agency (denormalised for routing), locale, source enum contact · property · landing · boat_filter · whatsapp · list_with_us, consentMarketing bool + consentedAt + consentIp, utm json, status enum new · sent · viewed · qualified · spam, navigationPath json (empty), crmContactId (empty).
- LandingPage, Destination, WaterBody, Taxonomy, Article, Page, Redirect, User, AuditLog** (who changed what, when — required for multi-agency trust), **ImportJob** (file, agency, status, rows processed/failed, error report).

6.9 Indexes (create explicitly)

slug · agency · status · moderation · visibility · isSample · featured · priceEur · propertyType · waterBodyType · waterFrontageM · maxBoatLoaM · waterDepthAtBerthM · navigableToOpenSea · country · destination · expiresAt · fingerprint · GIST on coordinates · composite (status, visibility, isSample) for the hot public query.

7. Stack and setup

7.1 Decisions

Area	Choice	Why
Framework	Next.js 15+ App Router, TS strict	SSG/ISR/PPR, image pipeline, edge CDN
CMS + backoffice	Payload 3 embedded in the same app	No second server, Postgres-native, generated types, real admin UI, granular access control — which is exactly what multi-agency needs
DB	Postgres 16 + PostGIS (Neon)	Geo queries must be DB-side; retrofitting PostGIS is painful
Search	Typesense Cloud	Sub-50 ms faceted search with counts, geo filters, typo tolerance. Postgres remains source of truth
Styling	Tailwind v4 + token CSS variables	One file restyles everything
Maps	MapLibre GL + MapTiler/Protomaps	Cheaper, faster, no lock-in, custom water style
Media	Cloudflare R2 + Cloudflare Images	Variants at upload, global CDN, predictable cost
i18n	next-intl	
Forms	Zod + React Hook Form	Shared schemas
Email	Resend + React Email templates	Leads, agency notifications
Queues/cron	Vercel Cron + a Job table	Feed imports, expiry sweeps, reindex
PDF	@react-pdf/renderer	No headless browser on serverless
Analytics	Plausible	Privacy-first, lightweight
Monitoring	Sentry + Vercel Speed Insights	Real-user CWV
Testing	Vitest + Playwright + axe-core + Lighthouse CI	
Deploy	Vercel	ISR, previews, edge cache

7.2 Bootstrap

```
pnpm create payload-app@latest waterline --template website --db postgres --use-pnpm
cd waterline

pnpm add next-intl zod react-hook-form @hookform/resolvers maplibre-gl typesense \
  @react-pdf/renderer resend react-email @react-email/components blurhash sharp \
  papaparse xml2js date-fns clsx tailwind-merge lucide-react @vercel/og

pnpm add -D tailwindcss @tailwindcss/postcss autoprefixer vitest @vitejs/plugin-react \
  playwright @playwright/test @axe-core/playwright eslint prettier \
  prettier-plugin-tailwindcss typescript @faker-js/faker tsx dotenv-cli \
  @next/bundle-analyzer @lhci/cli

docker run -d --name wl-pg -p 5432:5432 -e POSTGRES_PASSWORD=waterline \
  -e POSTGRES_DB=waterline postgres/postgis:16-3.4
docker run -d --name wl-ts -p 8108:8108 -v /tmp/typesense:/data \
  typesense/typesense:27.1 --data-dir /data --api-key=devkey --enable-cors

psql postgresql://postgres:waterline@localhost:5432/waterline \
  -c "CREATE EXTENSION IF NOT EXISTS postgis; CREATE EXTENSION IF NOT EXISTS pg_trgm;"

pnpm payload generate:types && pnpm dev
```

7.3 Environment variables

```
DATABASE_URL=
NEXT_PUBLIC_SITE_URL=
R2_ACCOUNT_ID= R2_ACCESS_KEY_ID= R2_SECRET_ACCESS_KEY= R2_BUCKET=
NEXT_PUBLIC_CF_IMAGES_URL=
TYPESENSE_HOST= TYPESENSE_API_KEY= TYPESENSE_SEARCH_ONLY_KEY=
RESEND_API_KEY= LEAD_NOTIFY_TO= AGENCY_NOTIFY_FROM=
FX_API_KEY=
UNSPLASH_ACCESS_KEY=
NEXT_PUBLIC_PLAUSIBLE_DOMAIN=
CRON_SECRET=
PAYLOAD_SECRET=
NEXT_PUBLIC_DEFAULT_LOCALE=en
NEXT_PUBLIC_MAPTILER_KEY=
SAMPLE_DATA_ENABLED=true
SENTRY_DSN=
FEED_INGEST_SECRET=
```

7.4 Folder structure

```
/src
/app/[locale]/...      public routes
/app/(payload)/...     admin
/app/api/...           health, og, revalidate, feeds, leads, brochure, cron
/components/{ui,property,search,layout,admin}
/lib/db                typed data access – the ONLY place touching the DB
/lib/search            Typesense client, schema, sync
/lib/geo               PostGIS helpers, bbox, jitter, distance
/lib/seo               combos resolver, sitemaps, JSON-LD, metadata helper
/lib/import            CSV parser, XML feed adapters, field mapping, validation
/lib/services          EMPTY – future CRM/AI boundary
/lib/pdf /lib/email /lib/schemas
/payload/collections /payload/access /payload/hooks
/tokens /messages /scripts
```

8. Multi-agency backoffice (the supply engine)

This section did not exist in v1.0 and is what turns the site from a demo into a portal. Payload's access-control system does the heavy lifting; the work is in designing the workflow correctly.

8.1 Roles and permission matrix

Capability	admin	editor	agency_admin	agency_agent
See all agencies' listings	✓	✓	✗ own agency only	✗ own listings only
Create/edit listing	✓	✓	✓ own agency	✓ own only
Publish directly	✓	✓	✗ → pending_review	✗ → pending_review
Approve / reject listings	✓	✓	✗	✗
Feature a listing	✓	✓	✗	✗
Manage agency profile & agents	✓	✗	✓ own	✗
Bulk import / configure feed	✓	✗	✓ own	✗
See leads	✓ all	✓ all	✓ own agency	✓ own listings
Edit landing pages, journal, taxonomy	✓	✓	✗	✗
View audit log	✓	✗	✓ own agency	✗
Manage users & roles	✓	✗	✓ own agents	✗

Implementation: a `tenant` access function in `/payload/access/tenant.ts` applied to every collection; **write one function and reuse it** — scattered per-collection logic is how tenant leaks happen. Add a Playwright test that logs in as Agency B and asserts a 403/empty result on Agency A's listing.

8.2 Agency onboarding

- Agency applies at `/list-with-us` (company, country, website, licence number, contact, inventory size, how many listings have direct water access).
- Admin reviews, creates the `Agency` record and an `agency_admin` user; Payload sends a set-password email.
- Agency completes its profile (logo, description, languages, contacts) — profile completeness is shown as a progress bar and gates the first publish.
- Agency adds agents, then uploads listings by one of three routes (§8.3–8.5).
- First five listings are reviewed manually; after a clean record, the agency can be promoted to `tier = verified`, which auto-approves listings that pass automated validation.

8.3 Route A — manual entry

Payload admin with a **custom listing form layout** grouped as: Essentials → **Water credentials** → Nautical → Physical → Location → Media → Content → Publishing. The water group sits second, not last, so it is never skipped. Live preview and a completeness meter ("Listings with frontage data receive 2.4× more enquiries — add it") are visible while editing.

8.4 Route B — bulk CSV/XLSX import

Downloadable template with a data-dictionary sheet. Column set (required marked *):

```
reference*, title_en*, description_en*, property_type*, status*,
price_type*, price_amount, currency*, tenure,
bedrooms*, bathrooms*, built_area_sqm*, plot_area_sqm, year_built, condition,
water_body_type*, water_body_name, water_access_types* (pipe-separated),
distance_to_water_m*, water_frontage_m, beach_type, orientation, swimmable,
shoreline_tenure, concession_expiry, flood_zone, waterfront_protection,
mooring_type, berth_count, max_boat_loa_m, max_boat_beam_m, water_depth_at_berth_m,
navigable_to_open_sea, fixed_bridges_to_open_sea, min_bridge_clearance_m,
nearest_marina_name, nearest_marina_distance_km,
country*, region, province, locality*, address_line, latitude*, longitude*,
coordinate_precision, features (pipe-separated), image_urls* (pipe-separated),
agent_email, video_url, virtual_tour_url
```

Import flow: upload → **dry-run validation report** (row-by-row: ok / warning / error with the exact reason and column) → the agency fixes and re-uploads or confirms → import runs as a background job → summary email with a downloadable error CSV. Imports are **idempotent on** (agency, reference). Images are fetched, validated (≥1600 px, not a stock watermark, not a duplicate hash), optimised and stored. An `ImportJob` record keeps the full audit.

8.5 Route C — XML/JSON feed

Documented inbound contract at `/api/feeds/[agencySlug]` with a per-agency token, supporting: our native JSON schema (published as a JSON Schema file at `/schemas/listing.schema.json`), plus adapters for the common shapes agencies already produce — Kyero XML, Resales-Online, Houzez/WP REST export, and a generic mapped CSV over HTTP. Feeds are polled by a Vercel Cron job every 6 h; each run produces a diff report (created/updated/withdrawn/errors) emailed to the agency and visible in the admin. Listings absent from two consecutive feed runs are moved to `withdrawn`, never deleted.

Field-mapping UI: for generic feeds, an admin screen maps source fields to our schema and stores the mapping on the `Agency`. Write it once; it removes 80 % of onboarding friction.

8.6 Moderation workflow

`draft` → `pending_review` → `approved (in_market)` | `changes_requested` | `rejected`.

The review queue is a custom admin page showing side-by-side: submitted data, automated validation results, image quality flags, duplicate matches, and a map pin check. Automated pre-checks: water rule, coordinate plausibility (point must be within 200 m of a water polygon — use a coarse coastline/water dataset, e.g. Natural Earth or OSM water polygons), price sanity versus destination median, minimum 6 images, description length ≥ 300 characters, no phone numbers or emails in the public description, no ALL-CAPS titles. Anything flagged goes to a human; everything clean from a `verified` agency auto-approves.

8.7 Listing freshness

`expiresAt = publishedAt + 180 days`. At T-14 days the agency receives "confirm these 12 listings are still available" with one-click confirmation per listing. On expiry the listing moves to `expired`: removed from search and sitemap, page returns **410 Gone** with a designed page offering similar properties (\$14.5). Stale inventory is the single biggest credibility killer in property portals.

8.8 Duplicate detection

Compute `fingerprint` from rounded coordinates (4 dp) + property type + built area bucket + bedroom count, plus perceptual hashes of the first three images. On a match: flag in the review queue, let the admin link `duplicateOf`, and show only the canonical listing publicly with "also listed by" agency attribution. Two agencies legitimately sharing a mandate is normal; two identical public pages is an SEO and trust problem.

8.9 Lead routing and SLA

Lead → saved → routed to `property.agent` if set, else the agency's default lead inbox, else the internal desk. Email plus optional WhatsApp deep link. Every lead has a status; if it stays `new` for 48 h the agency gets a reminder and the admin dashboard flags it. Response-time statistics per agency feed the `tier` decision. All lead emails are sent from our domain with reply-to the enquirer, so we retain the record.

8.10 Agency-facing dashboard

A custom admin home for `agency_admin`: listings by status, listings needing attention (missing frontage, expiring, changes requested), views and leads per listing over 30 days, average response time, feed health, and a "download your listing data" export. Give agencies numbers and they keep the inventory fresh without being chased.

9. Design system

9.1 Brand placeholder policy

The brand name appears in exactly one constant, `/src/config/brand.ts` (name, legalName, domain, tagline, email, phone, whatsapp, social handles). All UI, emails, PDFs, metadata and JSON-LD read from it. Renaming the product is a one-line change plus logo assets.

9.2 Tokens — `/src/tokens/tokens.ts`

```
export const tokens = {
  color: {
    ink: '#08141C', inkSoft: '#22323C', abyss: '#0B2A38', tide: '#12586B',
    surf: '#7FB6BE', sand: '#D9CBB6', shell: '#F6F3EE', white: '#FFFFFF',
    line: `rgba(8,20,28,0.12)`, focus: '#12586B',
    success: '#2E7D5B', warning: '#B07A24', danger: '#A33A2E',
  },
  font: {
    display: '"Canela","Playfair Display",ui-serif,Georgia,serif',
    body: '"Inter",ui-sans-serif,system-ui,sans-serif',
    mono: '"ui-monospace","SF Mono",monospace',
  },
  size: {xs: '0.75rem', sm: '0.875rem', base: '1rem', lg: '1.25rem', xl: '1.5rem',
    '2xl': '2rem', '3xl': '2.75rem', '4xl': '3.75rem', '5xl': '5rem'},
  space: [0, 4, 8, 12, 16, 24, 32, 48, 64, 96, 128],
  radius: {none: '0', sm: '2px', md: '4px', lg: '8px', pill: '999px'},
  shadow: {card: '0 1px 2px rgba(8,20,28,.06)', pop: '0 8px 32px rgba(8,20,28,.14)'},
  z: {map: 10, sticky: 20, header: 30, drawer: 40, modal: 50, toast: 60},
  bp: {sm: '640px', md: '768px', lg: '1024px', xl: '1280px', '2xl': '1536px'},
  motion: {fast: '160ms', base: '240ms', slow: '320ms', ease: 'cubic-bezier(.2,.6,.2,1)'},
} as const;
```

Acceptance: changing `color.abyss` and `font.display` restyles the whole site *and* the PDF brochure *and* the transactional emails, with no other edit.

9.3 Art-direction rules (enforced in components)

- One image per viewport minimum: heroes full-bleed, ≥ 72 vh desktop / 62 vh mobile.
- Every photo gets a gradient scrim `linear-gradient(180deg, rgba(8,20,28,0) 40%, rgba(8,20,28,.72) 100%)` — text stays legible on any photo.
- Fixed aspect ratios only: hero 21:9 / 4:5 mobile, cards 3:2, gallery 3:2, editorial 16:9. No intrinsic sizing anywhere → CLS 0 by construction.
- One uniform colour treatment on all listing photography (`saturate(1.04) contrast(1.03)`) so mixed-quality agency photos read as one brand.
- Type does the luxury work: display serif at large sizes with tight tracking, neutral grotesque elsewhere. No script fonts, no gold, no gradients.
- Motion: 160–320 ms fades plus 8 px translate on scroll-in, `prefers-reduced-motion` honoured, no parallax, no scroll-jacking.
- Fonts self-hosted, subset (Latin + Cyrillic for `ru`), `font-display: swap`, max 4 files, only the above-the-fold display weight preloaded.

9.4 Component inventory

Button · IconButton · Input · Select · Combobox (destination typeahead) · RangeInput · Checkbox · Switch · Chip/FilterPill · Tooltip · Dialog · Drawer · Tabs · Accordion · Container · Section · Header · Footer · LocaleSwitcher · CurrencySwitcher · UnitSwitcher · PropertyCard · PropertyCardCompact · Gallery · WaterCredentialsTable · NauticalPanel · BoatFitBadge · FilterBar · MapPanel · ResultsList · StatStrip · EditorialBlock · LeadForm · Breadcrumbs · Pagination · Skeleton · EmptyState · Toast · SampleBadge.

Every one is documented in a dev-only `/dev/styleguide` route with all states including focus, disabled, loading, error and RTL-safe spacing.

10. Page specifications

10.1 Home — SSG + ISR 300 s

- Hero** — full-bleed still (or ≤ 1.5 MB muted AV1 loop with poster), display headline, one line of subcopy, and the search bar: *Where* (typeahead over Destinations + WaterBodies) · *Water type* · *Price* · **Boat length (m)** · Search.
- Signature listings** — 6 featured cards showing price · location · **frontage** · water body · access icon.
- Browse by water** — 6 tiles: Sea · Lake · River & Canal · Lagoon · Fjord · Private Islands → landing pages.
- Destinations** — 8–12 tiles with live counts.

5. **"Will it fit?"** — boat-length slider with a live count of matching properties. The brand moment; counts for five bucket values are precomputed at build so it feels instant.
6. **Why WATERLINE** — three short proofs (verified water access, frontage data, berth data).
7. **Journal** — latest 3 articles.
8. **List with us** — supply-side CTA.
9. Footer with destination links, locale/currency/unit switchers, legal.

10.2 Search

Desktop 60/40 split (virtualised list left, sticky map right); mobile list with a floating Map toggle → full-screen map with a synced bottom card carousel.

Filters: price · water body type · **water access type** · **min frontage** · **boat LOA + draft** · navigable to open sea · no fixed bridges · property type · beds · baths · min built sqm · min plot sqm · orientation · beach type · tenure · country/destination · status. Live facet counts on every option.

Behaviours: search-as-I-move-the-map (default on, 350 ms debounce, bbox in URL) · marker ↔ card hover sync · clustering above 60 markers · circles for approximate-precision listings · sorts (featured, newest, price ±, **frontage desc**, **max boat length desc**) · computed empty state · shareable URL · back/forward restores state exactly.

Performance: static shell, Typesense for facets and ids, cached Postgres hydration, map dynamically imported (`ssr:false`) and never in the initial bundle.

10.3 Listing detail — SSG + ISR 600 s

Gallery (first image `priority`, rest lazy, lightbox code-split, keyboard + swipe, blurhash) → headline (title, location, price or "Price on request", reference, share, save-to-clipboard) → **Water credentials table** → **Nautical panel** (only when populated) with "Fits a XX m yacht" badge → key facts → highlights + description → map (lazy, precision-aware) → features grid → **similar properties** (`/lib/similar.ts`, swappable for embeddings later) → agency/agent card + inquiry form + WhatsApp → brochure PDF → breadcrumbs → JSON-LD (`RealEstateListing`, `Offer`, `BreadcrumbList`) → view counter (server-side, session-deduplicated, non-blocking).

States to design: sold (banner + "see similar"), under offer, expired (410 page), unlisted (noindex + "shared privately" note), sample (badge + banner).

10.4 Landing page

Editorial title + intro → **live stats strip** (count, median price, median frontage, most common type, price range — computed at render, never stored) → 12 matching listings + "view all" opening search with filters pre-applied → editorial body (area guide, mooring notes, buying notes) → **internal links block** (6–10 siblings) → FAQ accordion with `FAQPage` JSON-LD → lead form. Renders only when `published` and `intro` is non-empty; `generateStaticParams` returns published records only.

10.5 Others

Destination hub and destination pages · Collections ("Islands under €5 M", "Homes with a 30 m berth") · Journal index + article (with author, reading time, `Article` JSON-LD, related listings) · Agency profile pages (logo, description, verified badge, their listings — supply-side SEO and trust) · About · List with us (application form) · Contact · Legal · designed 404 and 500 with search and popular destinations.

11. Copy deck (use verbatim; translate for the other 5 locales)

The developer never needs to invent copy. Everything below goes into `/src/messages/en.json`.

11.1 Home

- **Hero H1:** "Property where the water begins."
- **Hero sub:** "Every listing has direct access to the sea, a lake, a river or a canal. Verified, measured, and searchable by the length of your boat."
- **Search CTA:** "Search waterfront"
- **Signature section:** "Signature properties" / "A selection from the current market."
- **Browse by water:** "Choose your water" / "Sea · Lake · River & canal · Lagoon · Fjord · Private islands"
- **Boat strip H2:** "Will it fit?" / "Set your boat's length and see only the properties with a berth that takes it." / Result: "{count} properties take a {loa} m boat."
- **Why section:** "Verified access — no sea views, no 'five minutes from the beach'." · "Measured frontage — we publish metres of private shoreline." · "Berth data — depth, length, bridges, and the route to open water."
- **Supply CTA:** "List with us" / "Free international exposure for properties that meet one condition: the water starts at the property line."

11.2 Listing page

- Price on request → "Price on request"
- Water block title: "Water credentials" · Nautical: "Berth & navigation"
- Labels: "Water body", "Access", "Private frontage", "Distance to water", "Beach", "Orientation", "Swimmable from the property", "Shoreline tenure", "Shoreline protection", "Mooring", "Berths", "Max boat length", "Max beam", "Depth at berth", "Navigable to open sea", "Fixed bridges", "Nearest marina"
- Badge: "Fits a {n} m yacht" · "Direct access to open sea" · "No fixed bridges"
- Enquiry: "Request details" / "Ask about this property — the agency replies directly." / Consent: "I agree that my details are shared with the listing agency and processed as described in the Privacy Policy."
- Similar: "Comparable waterfront properties"
- Sold: "This property has been sold." · Expired: "This listing is no longer available."
- Sample: "Sample listing — demonstration data, not a real property."

11.3 Search

- Empty: "No property matches all of your filters. The narrowest one is **{filter}** — relaxing it shows {count} properties." + button "Relax {filter}"
- Loading: skeletons only, no spinner text.
- Map toggle: "Map" / "List" · "Search as I move the map"

11.4 Meta templates (§14.2 implements these)

- Listing: {title} — {propertyType} with {accessType} in {locality}, {country} | {brand} · description: {bedrooms} bedrooms, {builtAreaSqm} m², {waterFrontageM} m of private {waterBodyType} frontage in {locality}. {priceOrOnRequest}.
- Landing: {PropertyTypePlural} with {access} in {Location} — {count} for sale | {brand} · description: Browse {count} {propertyTypePlural} with direct water access in {Location}. Median price {medianPrice}, median frontage {medianFrontage} m. Verified waterfront listings only.
- Destination: Waterfront property for sale in {Destination} | {brand}

11.5 Transactional emails (React Email templates)

Lead to agency · lead confirmation to enquirer · agency welcome + set password · import summary · feed diff report · listing approved / changes requested / rejected · expiry reminder (T-14) · lead-unanswered reminder (48 h) · weekly agency digest. Each: token-styled, plain-text fallback, one CTA, no images beyond the logo.

11.6 Legal pages (outline for the client's lawyer, drafted by the developer as placeholders)

Privacy policy (controller, data collected, lawful basis, recipients incl. agencies, retention §16.5, rights, DPO contact) · Cookie policy with the live cookie table · Terms of use (no brokerage, accuracy disclaimer, IP, prohibited scraping) · Agency terms (content ownership and licence to display, accuracy warranty, removal rights) · Listing disclaimer displayed on every property page: "Details are provided by the listing agency. Measurements, boundaries, berth dimensions and shoreline rights are indicative and must be verified independently before any transaction."

12. Performance — budgets and rules

12.1 Budgets (acceptance criteria; mobile, Moto G class, throttled 4G)

Metric	Target	Fail
LCP	< 1.2 s	> 1.8 s
CLS	< 0.02	> 0.05
INP	< 150 ms	> 200 ms
TTFB (edge-cached)	< 150 ms	> 400 ms
First-load JS — home	< 110 KB gz	> 150 KB
First-load JS — listing	< 130 KB gz	> 170 KB
First-load JS — search (pre-map)	< 160 KB gz	> 200 KB
Hero image transferred	< 120 KB AVIF	> 200 KB
Lighthouse mobile performance	≥ 95	< 90
Lighthouse accessibility / best practices / SEO	100 / ≥ 95 / 100	below
Third-party scripts before interaction	1 (analytics, deferred)	≥ 2
Typesense query p95	< 60 ms	> 150 ms

12.2 Rules

- Everything public is SSG or ISR; `force-dynamic` forbidden on indexable routes.
- Images: `next/image` or a Cloudflare loader, AVIF first, explicit `sizes` on every instance, variants at upload never on demand, exactly one `priority` image per page, blurhash everywhere else.
- JS discipline: map, lightbox, charts, PDF renderer all `next/dynamic` with `ssr:false`. No moment.js, no whole-lodash, no UI kit shipping more than it renders.
- Zero layout shift: fixed ratios, reserved space, skeletons matching final dimensions exactly.
- Caching: ISR + `stale-while-revalidate`, plus a signed `/api/revalidate` webhook from Payload `afterChange` so edits appear in seconds without dynamic rendering.
- DB: every filter field indexed, GIST on coordinates, materialised views for landing-page aggregates refreshed every 15 min once inventory passes 5,000.
- Third parties: analytics only, `afterInteractive`. Chat widgets, tag managers, pixels and A/B tools are **forbidden in Phase 1** — the classic 95 → 60 Lighthouse collapse.
- CI enforces budgets; a regression fails the build.
- Perceived speed: prefetch on card hover/viewport, optimistic filter chips, instant skeletons, View Transitions where supported.

13. Content sourcing and production — where everything comes from

This chapter exists so that the developer never has to ask "what do I put here?". Every asset the site needs is listed below with a named source, a licence position and a production method. **Nothing in this build waits on the client.**

13.1 Content inventory

Asset	Quantity for launch	Source	Effort
Listing photography	~600 images	Unsplash / Pexels API via the generator (\$13.2)	Automatic
Listing data	60 listings	Generator (\$13.10)	Automatic
Listing descriptions, 6 locales	360 texts	Template grammar (\$13.8)	Automatic
Landing-page editorial	40 × 350 words	AI-drafted, developer-reviewed (\$13.7)	~2 days
Journal articles	8 × 900 words	AI-drafted from named sources (\$13.7)	~1.5 days
Destination intros	12 × 200 words	Same	~0.5 day
UI strings, 6 locales	~450 keys	Copy deck \$11 + machine translation (\$13.9)	~0.5 day
Geodata (destinations, coordinates, polygons)	12 destinations, 40 water bodies	GeoNames + Natural Earth + OSM (\$13.5)	~0.5 day
Fonts	2 families	Google Fonts, self-hosted (\$13.3)	1 hour
Icons	~40	Lucide (\$13.3)	—
Logo / wordmark	1 lockup + favicon set	Type-only recipe (\$13.4)	2 hours
Agency & agent identities	6 + 12	Fictional, generated (\$13.2.4)	Automatic
Legal pages	5	Generator templates, marked as placeholders (\$13.11)	2 hours
Emails	10 templates	Copy deck \$11.5 + React Email	~1 day
FX rates	daily	ECB via Frankfurter API (\$13.6)	1 hour

Total content effort: roughly **one week inside the build**, with zero client input.

13.2 Photography

Free, commercially usable, API-accessible:

Source	API	Licence position	Use for
Unsplash	unsplash.com/documentation	Free for commercial use; attribution not legally required but the download-tracking call is required by their API terms	Primary source for all listing and editorial imagery
Pexels	pexels.com/api	Free commercial use, no attribution required	Fallback and variety
Pixabay	pixabay.com/api/docs	Free commercial use	Third fallback
Openverse	api.openverse.org	Aggregates CC-licensed work — check each licence, some require attribution or forbid commercial use	Only when the others fail
Wikimedia Commons	commons.wikimedia.org/w/api.php	Mixed licences, often attribution + share-alike	Editorial illustration of named places only

Paid, if uniqueness matters later: Unsplash+ (~€10/month, no attribution, better exclusivity), Adobe Stock, Getty/iStock, Stocksy (highest quality, highest price), Shutterstock. A single Adobe Stock 10-image pack is enough to upgrade the six homepage hero slots if the free imagery ever looks generic.

13.2.1 Query sets. Store them in `/src/scripts/image-queries.ts` as a typed map so they can be tuned without touching the generator:

```
export const imageQueries = {
  byDestination: {
    liguria: ['portofino coast villa', 'italian riviera sea', 'ligurian coastline house'],
    lake_como: ['lake como villa', 'como lakefront terrace', 'italian lake boathouse'],
    amalfi: ['amalfi coast villa', 'positano cliff house', 'mediterranean terrace sea'],
    cote_dazur: ['french riviera villa', 'cap ferrat sea view', 'mediterranean modern villa'],
    florida_keys: ['florida waterfront mansion', 'miami canal house dock', 'key west waterfront'],
    hamptons: ['hamptons beach house', 'long island waterfront home'],
    norway: ['norway fjord house', 'boathouse fjord', 'scandinavian waterfront cabin'],
    netherlands: ['amsterdam canal house', 'dutch canal waterfront'],
    // ...one entry per destination
  },
  byFeature: {
    private_dock: ['private dock boat house', 'wooden jetty lake', 'yacht berth private'],
    private_beach: ['private beach villa', 'white sand beach house'],
    private_island: ['private island aerial', 'small tropical island'],
    interior: ['luxury living room sea view', 'modern kitchen ocean view', 'master bedroom sea'],
    pool: ['infinity pool ocean', 'pool terrace sunset sea'],
    aerial: ['aerial coastline villa', 'drone shot waterfront property'],
  },
} as const;
```

13.2.2 Selection rules (implement as filters, not as taste). Reject any image that: is under 1600 px on the long edge; carries a visible watermark or stock overlay; contains a recognisable face in the foreground; shows a recognisable brand, logo or licence plate; is a known landmark used as a stand-in for a private home; duplicates a perceptual hash already used in the same destination. Prefer landscape 3:2, daylight, no heavy filters. Compose each listing gallery as: 1 exterior hero, 2 exterior/water, 3-5 interior, 1 aerial, 1 detail — so galleries feel authored rather than random.

13.2.3 Storage discipline. Always persist `credit`, `licence`, `sourceUrl` and `sourceId` on `Media`. Render attribution in the gallery footer of sample listings. This makes the whole demo defensible in one line if anyone asks.

13.2.4 People imagery — a real rule, not a nicety. Never use a stock portrait of a real person as a fictional agent. It creates an identifiable person apparently endorsing a business they have never heard of, which is a personality-rights and GDPR problem independent of the stock licence. For sample agents use: monogram avatars generated from initials in brand colours (a 20-line SVG component), or silhouette placeholders. Real photographs only for real, consenting staff.

13.2.5 AI-generated imagery. Acceptable for abstract editorial headers and texture, using a model whose terms permit commercial use (Adobe Firefly is the safest on training provenance; Flux and Midjourney are usable under their commercial tiers). **Not acceptable for listing photography**, even in demo data: an invented photorealistic house presented as a property is exactly the misrepresentation §13.12 exists to prevent. If used anywhere, record `licence: 'ai-generated'` and the model name on the `Media` record.

13.3 Fonts and icons

Recommended pairing (free, self-hosted from Google Fonts): display `Playfair Display` or `Cormorant Garamond`; body `Inter`. Alternatives with more character: `Fraunces` (display), `Instrument Serif` (display), `Libre Caslon Display`, `Newsreader`; body alternatives `Geist`, `Public Sans`, `Source Sans 3`.

Paid upgrades if the brand budget appears later: Canela, GT Sectra, Ogg, Tiempos (display); Söhne, Neue Haas Grotesk, ABC Diatype (body). Licensing is per-domain and per-pageview tier — note it in `DECISIONS.md` and never self-host a paid font without the web licence.

Download with `google-webfonts-helper` or `fontsource`, subset to Latin + Latin-Extended + Cyrillic with `glyphhanger`, serve WOFF2 only, max four files total.

Icons: Lucide (already a dependency, ISC licence). Phosphor as an alternative. Draw the five water-access icons (beach, dock, mooring, boathouse, island) as bespoke 24 px SVGs on Lucide's grid — five icons is an afternoon and it is the visual signature of the product.

13.4 Logo and favicon without a designer

A type-only wordmark is correct for this category and takes two hours:

- 1. Set the brand name in the display font, all caps, `letter-spacing: 0.18em`, weight 400, optical size ≥ 48 px.
- 2. Lockup: wordmark alone; optional 1 px rule above it at 60 % of the wordmark width (the horizon motif from §9.3).
- 3. Monogram for the favicon and app icons: first letter in the display face on `color.abyss`, 12 % padding.
- 4. Export SVG + PNG at 1x/2x, then generate the full icon set with `realfavicongenerator.net` and commit `site.webmanifest`.
- 5. Store everything in `/public/brand/` and reference it only through `/src/config/brand.ts`.

13.5 Geodata — destinations, coordinates, water bodies

Need	Source	Licence
Cities, provinces, regions with coordinates, population and localised alternate names	GeoNames (<code>geonames.org</code> , full dump or API)	CC BY 4.0 — attribute in the colophon
Country and region polygons for map context	Natural Earth (<code>naturalearthdata.com</code>)	Public domain
Coastlines, water polygons, marinas, piers, slipways	OpenStreetMap via Overpass API (<code>overpass-turbo.eu</code>)	ODbL — attribute "© OpenStreetMap contributors" on every map
Geocoding an address during import	Nominatim (respect the 1 req/s policy) or a paid geocoder for bulk	ODbL
Localised country and region names	<code>i18n-iso-countries</code> + CLDR	Open
Map tiles	MapTiler (free tier to 100k loads) or self-hosted Protomaps <code>.pmtiles</code>	Commercial / BSD

Use OSM properly, because it is a competitive advantage here. An Overpass query for `harbour`, `marina`, `pier`, `slipway` and `mooring` near each destination gives us a real marina dataset for the `nearestMarina` fields and for the coordinate-plausibility check in the moderation queue (§8.6). Cache the extract in `/data/osm/` and refresh quarterly; do not query Overpass at runtime.

13.6 Rates, units and reference data

FX: **Frankfurter API** (`api.frankfurter.app`, free, ECB reference rates) or `exchangerate.host`. Fetch once a day via cron, store a `FxSnapshot` row, and always convert from the stored snapshot so prices never change mid-session. Units, number and date formatting come from `Intl`; never hand-roll them.

13.7 Editorial copy — the writing brief and the AI workflow

The developer writes the editorial with AI assistance, following these rules. This is a legitimate, high-quality workflow; unreviewed AI output is not.

Structure for every landing page (350–450 words): a 40–60-word direct answer paragraph opening the page · what the market is like in one paragraph · what buyers actually get for the money, with real figures from our own aggregates · one paragraph of practical specifics (mooring rules, access, seasonality, tenure quirks) · a closing line pointing to the listings. Two question-form H2s. No adjectives that could apply to any coast.

Structure for a journal article (800–1,000 words): answer-first opening, three to five H2s in question form, one table or list of concrete figures, every non-obvious claim sourced with a link and a date, a short "what this means for buyers" close.

Four rules, non-negotiable:

1. Never publish a statistic the assistant produced from memory. Every number comes from a named source with a URL and a date, or from our own database aggregates.
2. Every external claim carries an inline source and `asOfDate`.
3. The developer reads every published sentence. Anything that cannot be verified in 30 seconds is deleted rather than softened.
4. No AI-generated text about law, tax or residency status beyond describing that a rule exists and linking the official source.

Article backlog for launch (pick 8): mooring and anchoring rules by country · buying a *demanio marittimo* concession in Italy, explained · what "no fixed bridges" means and why Florida buyers pay for it · dredging, draft and why depth at the dock matters · riparian rights versus foreshore ownership in England, France and Italy · flood zones and coastal insurance · what a metre of private frontage is actually worth · the ten best superyacht-capable private docks in the Mediterranean · lake versus sea: the trade-offs nobody explains · buying a private island: the eight questions to ask first · canal houses in Amsterdam: what the deeds actually say · winterising a waterfront property.

Named sources to draw on (cite, never copy): Knight Frank and Savills research libraries, Eurostat and national statistics offices, Agenzia delle Entrate OMI quotations for Italy, HM Land Registry Price Paid data for the UK, Notaires de France, US county assessor data and FRED, the BIS residential property price database, national maritime authorities for mooring rules. Quoting a sentence with attribution is fine; reproducing a table or a report section is not.

13.8 Listing descriptions — template grammar

Listing copy is generated, not written, from a slot grammar per locale in `/src/content/description-templates/{locale}.ts`:

```
{OpeningByType} {Positioning} {WaterSentence} {AccommodationSentence}
{OutdoorSentence} {NauticalSentence?} {LocationSentence} {ClosingByTier}
```

Each slot holds 6–10 variants selected deterministically by listing id, so no two listings read alike and re-running the generator is stable. Example water sentences: "The property holds {frontage} metres of private {waterBody} frontage, with {beachType} shoreline reached directly from the garden." Numbers always come from the structured fields, never from prose. This yields plausible, locale-native copy at zero marginal cost and is also exactly how real agency descriptions get written.

13.9 Translations

English is the source of truth. Draft the other five locales with the DeepL API (best quality for it/fr/de/es) or an LLM pass, then mark each translated string `machineTranslated: true` in the message file. Rules: never machine-translate legal pages or the NDA-equivalent terms; always hand-check the 40 highest-traffic strings (navigation, CTAs, form labels, error messages); keep a `/docs/translation-review.md` checklist for a native reviewer to clear later. Budget for a professional review of ~450 UI strings is roughly €300–500 per language — worth proposing to the client after launch, not before.

13.10 Sample inventory generator

`/src/scripts/seed-sample-data.ts` creates **60 properties** across 12 destinations (Liguria & Portofino, Lake Como, Amalfi Coast, Côte d'Azur, Mallorca, Ibiza, Dalmatia, Greek Islands, Algarve, Florida Keys & Miami, the Hamptons, Turks & Caicos) plus Norwegian fjord and Dutch canal outliers, with: internally consistent economics (price correlated with destination, type, area and frontage); water and nautical blocks populated with enough variety that **every filter returns at least three results**; 8–14 images each from §13.2; descriptions in all 6 locales from §13.8; 6 fictional agencies and 12 agents with monogram avatars; 10 `LandingPage` records; 8 journal articles.

13.11 Legal and policy text

Draft placeholders from a reputable generator (Termly, iubenda, or the GDPR.eu templates) plus the cookie table generated from the actual cookies in use, clearly marked `PLACEHOLDER – pending legal review` in a visible admin note and in a comment at the top of each file. The developer produces: privacy policy, cookie policy, terms of use, agency terms, listing disclaimer (§11.6). Legal review is a client task at the very end and does not block any development.

13.12 Sample-data safety rules (mandatory)

Publishing invented property adverts as real is a consumer-protection problem in most of our markets and poisons the index before launch. Therefore: every sample record has `isSample = true`; a visible "SAMPLE LISTING — demonstration only" badge and banner in all locales; `noindex`, `nofollow` and excluded from sitemaps while `SAMPLE_DATA_ENABLED=true`; contact forms log but never email an agent; no real agency, agent, address or phone number; monogram avatars only, never stock portraits; coordinates on plausible shoreline but never on an identifiable private home; `pnpm sample:purge` removes every sample record and its media; the launch checklist blocks go-live while any sample listing is publicly indexable.

13.13 Pre-launch content checklist

Every row must be green before DNS cutover: hero imagery selected and under budget · all 60 sample listings have ≥ 8 images and a full description in all locales · 40 landing pages have reviewed editorial copy · 8 articles published with sources · all UI strings translated and the top 40 hand-checked · destination and water-body data loaded from GeoNames/OSM with attribution rendered · fonts subset and self-hosted · logo and favicon set committed · legal placeholders in place and flagged · FX snapshot job running · sample data purged if the real inventory has arrived.

14. SEO and AI discovery

14.1 Foundations

Canonical on every page · hreflang for all 6 locales + `x-default` · sitemap index split by type with 50,000-URL / 50 MB children · `robots.txt` · designed 404/410 · one H1 per page · breadcrumbs everywhere · internal linking density of 6–10 contextual links per landing page · image filenames and alt text generated from structured data (`villa-private-dock-portofino-liguria-01.avif`, alt "Villa with private dock and 38 m of sea frontage in Portofino, Liguria").

14.2 Metadata

One helper builds every page's metadata from templates in §11.4, with fallbacks: `metaTitle || generated` and `metaDescription || generated from structured fields`. Titles ≤ 60 characters, descriptions 140–160. Never ship an empty description; never duplicate a title across combinations.

14.3 Structured data

`RealEstateListing` + `Offer` (or `PriceSpecification` for POA) with `numberOfRooms`, `floorSize`, `geo`, `amenityFeature` (map our water and nautical fields into `LocationFeatureSpecification` — this is how a machine learns we have berth data) · `BreadcrumbList` everywhere · `Organization` + `WebSite` with `SearchAction` · `FAQPage` on landing pages · `Article` on journal posts · `RealEstateAgent` on agency profiles. A Vitest test validates each template's shape; CI runs a Rich Results check on sample URLs.

14.4 Programmatic SEO

Phase 1 ships **~40 landing pages**, hand-seeded from the highest-intent combinations, each with real editorial copy:

```
villas-sea-liguria · lakefront-villas-como · seafront-villas-amalfi-coast
private-dock-homes-florida-keys · waterfront-homes-hamptons
beachfront-villas-mallorca · private-beach-villas-ibiza
private-islands-caribbean · private-islands-greece
canal-houses-amsterdam · fjord-properties-norway
marina-residences-dubai · waterfront-estates-cote-dazur
riverfront-homes-thames · lakefront-chalets-geneva ...
```

Post-launch expansion follows `waterAccessType × propertyType × destination` in batches, **each gated on written editorial copy**. Never mass-publish empty combinations. Internal linking: each landing page links to 6–10 siblings, each listing links to its 2–3 parent landing pages, the footer carries a curated destination block.

14.5 Lifecycle and crawl hygiene

- Sold: keep the page live for 90 days with a "sold" banner and links to similar, `noindex` after 30 days, then 301 to the parent landing page.
- Expired/withdrawn: **410 Gone** with a designed page (a 404 invites re-crawling; 410 removes it faster).
- Slug change: automatic `Redirect` record, 301, old URL never reused.
- Paginated search: `noindex`, `follow` beyond page 1, canonical to page 1.
- Filter URLs are never linked from crawlable navigation — only landing pages are.
- Weekly `pnpm audit:seo` reports: pages missing canonical/description, orphan pages, sitemap entries returning non-200, duplicate titles, listings with no inbound internal link.

14.6 AI and answer-engine discovery (new in v2.0)

Buyers increasingly start in an assistant, not a search box. Being answerable by machines is now part of SEO.

1. `/llms.txt` — a short Markdown map of the site for LLM agents: what the site is, the admission rule, the key sections, the landing-page index, the data licence, and contact. Mirror a fuller `/llms-full.txt` with the destination and taxonomy index.
 2. **Explicit crawler policy in `robots.txt`** — allow `GPTBot`, `0AI-SearchBot`, `PerplexityBot`, `ClaudeBot`, `Google-Extended` for public content; disallow `/admin`, `/api`, search-filter URLs and any `unlisted` content. Decide once, document the reasoning in `DECISIONS.md` (default: allow — discovery is worth more than the content).
 3. **Answerable content shape** — every landing page and article opens with a 40–60-word direct answer paragraph before any preamble, uses question-form H2s, and includes a facts block that machines can lift (counts, medians, ranges with units and dates). This is what gets quoted.
 4. **Entity clarity** — consistent `Organization` JSON-LD with `sameAs` links, an About page stating who we are in plain terms, and consistent naming of destinations and water bodies so the entity graph resolves.
 5. **Machine-readable data** — public JSON endpoints for aggregates (`/api/public/stats/[combo]`) returning counts and medians with an `updatedAt`; a JSON Schema for our listing shape. Cheap to build, disproportionately citable.
 6. **Freshness signals** — `dateModified` in JSON-LD and sitemaps, genuinely updated when data changes.
 7. **Editorial that answers real questions** — mooring rules by country, buying a *demanio marittimo* concession in Italy, "no fixed bridges" explained, dredging and draft, flood insurance for coastal buyers, the ten best superyacht-capable private docks. This ranks, proves authority, and is exactly the material assistants cite.
 8. **Measure it** — track referrals from assistant domains in Plausible and record monthly "does the assistant know us" spot checks in `DECISIONS.md`.
-

15. Accessibility — WCAG 2.2 AA (required, not optional)

Wealthy buyers skew older; accessibility here is commercial, not just legal (and in the EU the European Accessibility Act now applies to e-commerce-like services).

- Contrast $\geq 4.5:1$ body, $\geq 3:1$ large text and UI borders — verify the sand and surf tokens against `shell` before use.
 - Every interactive element is a real `<button>` or `<a>`, keyboard-reachable, with a visible 2 px `focus-visible` ring in `color.focus`. Never a `div` with `onClick`.
 - Gallery: arrow-key navigation, `Esc` closes, focus trapped in the lightbox and restored on close, image counter announced.
 - Map: a "skip map" link, and every map interaction has a non-map equivalent (the list is the source of truth).
 - Filters: labelled inputs, `aria-live` result count, no reliance on colour alone.
 - Forms: labels (not placeholders), errors linked with `aria-describedby`, error summary focused on submit.
 - Respect `prefers-reduced-motion`; target size $\geq 24 \times 24$ px (WCAG 2.2); no keyboard traps; logical heading order.
 - Automated axe-core checks on 8 routes in CI, plus one manual keyboard-only pass per major page recorded in `DECISIONS.md`.
-

16. Security, privacy, compliance

16.1 Application security

Payload admin behind strong auth, 2FA required for `admin` and `agency_admin`; server-side rate limiting on every public POST (leads 5/IP/h, feeds by token, search 60/min); CSP, HSTS, `X-Content-Type-Options`, `Referrer-Policy`, `Permissions-Policy`; no secrets in client bundles; signed webhooks (`REVALIDATE_SECRET`, `CRON_SECRET`, `FEED_INGEST_SECRET`); dependency audit in CI; Sentry with PII scrubbing.

16.2 Multi-tenant security

A single `tenant` access function reused across collections; a Playwright test per collection asserting cross-agency access fails; `AuditLog` on create/update/delete/publish/lead-view; uploaded media scoped to the owning agency; no agency can read another's leads, even via the API.

16.3 Anti-scraping

Our listing data is the asset. Rate-limit and bot-challenge aggressive clients (Cloudflare rules), omit exact coordinates for `approximate_500m`, never expose a public bulk JSON of all listings, watermark nothing but keep image URLs unguessable, and state the prohibition in the Terms. Accept that determined scraping cannot be stopped — the goal is to make it costlier than it is worth without harming legitimate crawlers.

16.4 GDPR

Non-blocking, deferred, granular cookie consent with a stored consent record; analytics only after consent where required; privacy, cookie and terms pages live before launch; DPAs with Vercel, Neon, Cloudflare, Resend, Plausible, Typesense, Sentry; data-processing register; a documented subject-access and erasure procedure with an admin action that anonymises a lead in one click.

16.5 Retention

Leads: 24 months, then anonymised. Import job files: 90 days. Audit log: 24 months. Analytics: aggregated only. Sold listings: retained indefinitely as market data but with enquirer data purged.

17. Analytics event schema

Plausible custom events, defined once in `/src/lib/analytics.ts`: `search_performed` (facets used, result count) · `filter_applied` (name, value) · `boat_filter_used` (loa, draft, results) · `map_moved` · `listing_viewed` (id, destination, priceBand) · `gallery_opened` · `brochure_downloaded` · `lead_submitted` (source, hasPhone) · `whatsapp_clicked` · `landing_viewed` (combo) · `agency_application_started/completed` · `locale_switched` · `currency_switched`. No personal data in event payloads. These events are the Phase 2 product-analytics foundation.

18. Testing

Unit (Vitest): filter → Typesense query translation (especially nautical), price FX normalisation, combo resolver and canonicalisation, metadata and JSON-LD builders, CSV row validation, duplicate fingerprinting, coordinate jitter determinism.

E2E (Playwright), critical paths only: home renders and LCP is the hero image · each filter returns correct results · boat LOA + draft excludes non-matching listings · map bbox updates results and URL · back/forward restores state · listing page renders all blocks with valid JSON-LD · lead submits, persists, routes and emails · unpublished landing page 404s · expired listing returns 410 · sample listings emit noindex and are absent from sitemaps · agency A cannot see agency B's listings or leads · CSV import dry-run reports the right errors · locale switch preserves path and query.

Non-functional: Lighthouse CI budgets, axe-core on 8 routes, bundle-size assertions, visual-regression screenshots on 10 viewports.

19. Runbooks

Deploy: merge to `main` → Vercel build → CI gates → automatic promotion. Rollback is a Vercel instant rollback plus, if the DB changed, the down-migration committed with every up-migration.

Database: Neon with daily backups and 7-day PITR; migrations are always reviewed as SQL before applying; never run a destructive migration without a verified backup timestamp recorded in `DECISIONS.md`.

Search index: `pnpm search:reindex` rebuilds into an alias, then swaps — never reindex in place. If Typesense is unreachable, search falls back to a Postgres query path with reduced faceting; this fallback must be implemented, not hypothetical.

Incidents: Sentry alert → check Vercel status and DB → roll back last deploy if the error started with it → post a note in `DECISIONS.md`. Availability target 99.9 %.

Content freeze / launch: 48 h before cutover, freeze agency uploads, run the final import, verify sitemaps and redirects, purge sample data, flip `SAMPLE_DATA_ENABLED=false`, then cut DNS with a 300 s TTL set 24 h earlier.

20. Definition of Done and launch checklist

1. All §12.1 budgets verified on PageSpeed Insights and in Lighthouse CI.
2. Every filter, including nautical, returns correct results with facet counts and map sync.
3. Multi-agency backoffice complete: roles enforced and tested, manual entry, CSV import with dry-run, at least one feed adapter working end-to-end, moderation queue, expiry sweep, lead routing.
4. 60 sample listings live, badged, `noindex`, out of sitemaps; `sample:purge` verified; the §13.13 content checklist is green.
5. 6 locales complete with hreflang and a working switcher.
6. ~40 landing pages published with real copy; unpublished combos 404; `llms.txt` served.
7. Valid JSON-LD on all templates; sitemaps submitted; `audit:seo` clean.
8. WCAG 2.2 AA: axe clean on 8 routes, keyboard pass recorded.
9. Leads save with consent, route correctly, notify, and are rate-limited.
0. An editor can publish a listing end-to-end with no developer involvement.
1. Changing `/tokens/tokens.ts` restyles site, PDF and emails.
2. `/lib/services` empty and documented; Phase 2/3 seams in place.

3. Security headers, 2FA, backups, Sentry, uptime monitor live.
4. `DECISIONS.md` complete, `README.md` explains setup in under 10 minutes for a new developer.
5. Staging signed off; DNS cutover checklist executed; 404/410 and ranking monitoring live.

21. Pre-answered decisions (so the client is not consulted)

Defaults are chosen to be safe and reversible. The developer follows them and logs any deviation.

#	Question	Default
1	Admission threshold	Direct water access only, ≤50 m, no price floor
2	Launch locales	en, it, fr, de, es, ru
3	Default currency	EUR, with USD/GBP/CHF/AED/SGD display conversion
4	Units	Metric default; imperial toggle for en-US/en-GB, stored in a cookie
5	Are prices required?	No — "Price on request" permitted, but such listings rank lower in default sort
6	Off-market handling	visibility = unlisted: viewable by link, noindex, excluded from sitemap
7	Exact coordinates	Agency chooses; default <code>approximate_500m</code> for listings above €5 M
8	Free or paid listings	Free in Phase 1; billing fields exist but are unused
9	Who can publish directly	Only admin/editor until an agency reaches <code>verified</code> tier
10	Listing expiry	180 days, reminder at T-14, then 410
11	Minimum images	6 to publish; 1600 px minimum long edge
12	Description minimum	300 characters, no phone numbers or emails in public text
13	Duplicate listings	Canonical listing displayed, others linked as "also listed by"
14	Lead ownership	Shared: agency receives it, we retain the record
15	Lead SLA	48 h, then reminder; response time feeds agency tier
16	Agency branding on listings	Logo + name + verified badge, no custom colours
17	Blog authorship	House byline "WATERLINE Editorial" until real authors exist
18	Sample data after launch	Purged before DNS cutover
19	AI crawlers	Allowed for public content; disallowed for admin/API/filters
20	Cookie banner style	Non-blocking bottom bar, reject-all in one click
21	Analytics vendor	Plausible, self-serve, EU-hosted
22	Newsletter	Capture only (double opt-in), no sending infrastructure in Phase 1
23	User accounts	None in Phase 1
24	Saved searches / alerts	Phase 2
25	Map provider	MapTiler; Protomaps self-hosted if tile cost exceeds €200/month
26	Map style	Custom: water saturated, land desaturated, minimal labels
27	Clustering threshold	60 markers
28	Default sort	Featured, then newest
29	Featured slots	6 on home, editorial choice only, max 2 per agency
30	Price bands for similar listings	±35 %
31	Pagination	24 results per page, infinite scroll on mobile with a URL-synced page param
32	Image treatment	Uniform filter profile; no watermarks in Phase 1
33	Video	Optional, YouTube/Vimeo embed loaded on click only (facade pattern)
34	Virtual tours	Link out, never embedded above the fold
35	Brochure PDF	Public on every listing, localised
36	WhatsApp	Shown when the agency provides a number

37	Phone numbers	Agency-level only, never per-listing free text
38	Error pages	Designed 404 and 410 with search and popular destinations
39	Domain strategy	Single domain, locale paths (not ccTLDs, not subdomains)
40	www vs apex	Apex canonical, www 301s
41	Trailing slash	No trailing slash, enforced with a redirect
42	Search engine for site search	Typesense; Postgres fallback path implemented
43	Uptime target	99.9 %, monitored externally
44	Support email	hello@{domain}, agencies@{domain}, privacy@{domain}
45	Handover artefacts	README, DECISIONS.md, env template, admin walkthrough video, launch checklist

22. READY-TO-PASTE CURSOR PROMPTS

Use in order. Each ends with acceptance criteria — require Claude to report pass/fail per line before you commit. Keep `CLAUDE.md`, §6 and §12 open as context.

Prompt 1 — Scaffold and guardrails

```
Set up the project "waterline":
- Next.js 15 App Router + TypeScript strict (noUncheckedIndexedAccess), Payload 3 embedded in the
  same app with the Postgres adapter, Tailwind v4 reading CSS variables, ESLint + Prettier.
- Folder structure exactly as spec §7.4. /src/lib/services contains only an empty index.ts.
- A zod-validated env module that fails fast at boot; .env.example with every var from §7.3.
- /src/config/brand.ts holding name, legalName, domain, tagline, email, phone, whatsapp, socials –
  every UI, email, PDF and JSON-LD reads the brand from here.
- /api/health returning { ok, db, search, version }.
- docker-compose.yml with postgres/postgis:16-3.4 and typesense:27.1; a migration enabling postgis
  and pg_trgm.
- Vitest + Playwright + axe + Lighthouse CI configured; GitHub Actions running typecheck, lint,
  test, e2e, lhci and bundle-size assertions on every PR.
- CLAUDE.md with the content from spec §1.1, /.cursor/rules/waterline.mdc, and an empty DECISIONS.md.
- package.json scripts: dev build start typecheck lint test test:e2e analyze seed sample:purge
  search:reindex audit:seo audit:ally audit:all.
ACCEPTANCE: pnpm dev boots, /api/health returns ok, pnpm typecheck/lint/test pass, CI is green,
and a deliberate lint violation fails the build.
```

Prompt 2 — Tokens and base components

```
Build the design system per spec §9.
1. /src/tokens/tokens.ts with the exact values in §9.2, frozen and typed.
2. Generate CSS custom properties from tokens and wire Tailwind v4 via @theme. Add an ESLint rule
  banning hex colours, px font-sizes and arbitrary spacing outside /src/tokens.
3. Self-host Inter + a serif display face, subset Latin+Cyrillic, font-display swap, preload only
  the above-the-fold display weight, max 4 files.
4. Build the component inventory in §9.4 using tokens only. Every interactive component must be a
  real button/anchor, keyboard-operable, with a visible focus-visible ring, and pass axe.
5. Fixed aspect-ratio wrappers for all image components.
6. /dev/styleguide route (dev-only) rendering every component in every state.
ACCEPTANCE: changing color.abysm and font.display restyles the whole styleguide; axe reports zero
violations on /dev/styleguide; tabbing reaches every control with a visible ring.
```

Prompt 3 — Property collection

```
Create the Payload collection Property exactly per spec §6.1–6.7, localised into en (default), it,
fr, de, es, ru. Include ALL fields listed there, including the multi-tenancy fields (agency, agent,
moderation, moderationNote, visibility, isSample, publishedAt, expiresAt, lastVerifiedAt,
duplicateOf, fingerprint, sourceType, viewCount, leadCount) and the full water and nautical blocks.

Hooks:
- beforeValidate: reject publish when waterAccessType is empty or distanceToWaterM > 50, with a
  human-readable reason stored in moderationNote and returned to the submitter.
- beforeChange: compute priceEur from a cached daily FX snapshot; generate slug on first publish
  only; compute fingerprint per §6.9/§8.8; set expiresAt = publishedAt + 180 days.
- afterChange: enqueue Typesense upsert and call the signed revalidation webhook.
- afterDelete: remove from Typesense.
Admin UI: field groups ordered Essentials → Water credentials → Nautical → Physical → Location →
Media → Content → Publishing; a completeness meter; a warning when waterFrontageM is null.
Enable drafts + versions + autosave. Create all indexes in §6.9 including GIST on coordinates.
Generate TS types. Show me the SQL migration before applying.
ACCEPTANCE: a listing without waterAccessType cannot be published; a listing with
distanceToWaterM=51 cannot be published; priceEur is correct for a USD listing; types compile;
EXPLAIN shows index usage on a filtered query.
```

Prompt 4 — Remaining collections and access control

Create the remaining collections per spec §6.8: Media (variants at upload per §12.2 rule 2, minimum 1600px long edge enforced with a clear error, blurbhash, credit/licence/sourceUrl, agency scoping), Agency, Agent, Lead, LandingPage, Destination, WaterBody, Taxonomy, Article, Page, Redirect, User, AuditLog, ImportJob.

Then implement multi-tenancy per spec §8.1:

- ONE reusable access function in `/src/payload/access/tenant.ts` used by every collection.
- Roles admin, editor, agency admin, agency agent with exactly the permission matrix in §8.1.
- agency admin and agency agent can never set `status=in_market` directly; saving sends the listing to pending review.
- AuditLog entries on create/update/delete/publish and on lead views.

ACCEPTANCE: Playwright tests prove agency B gets 403/empty on agency A's listings, media and leads, via both the admin UI and the REST/GraphQL API; an agency agent cannot publish; an editor can.

Prompt 5 — Data layer, geo and search index

Build the typed data layer; no route or component may query the DB directly.

1. `/src/lib/db`: `getPropertyBySlug`, `getFeatured`, `getSimilar`, `searchProperties(filters)`, `getAggregatesForScope`, `getDestinationCounts`, `getAgencyDashboardStats` – all with explicit types and all applying the public predicate (`status in_market/under_offer`, `visibility public`, `moderation approved`, `isSample` respecting `SAMPLE_DATA_ENABLED`).
2. `/src/lib/geo`: `bboxToPolygon`, `radiusQuery`, `ST_DWithin` helpers, and `jitterCoordinates` seeded by listing id so an approximate point never moves between requests.
3. `/src/lib/search`: Typesense schema with every facet in §10.2 plus a geo field; full reindex into an alias with atomic swap; incremental upsert/delete from the Property hooks; a Postgres fallback query path used automatically when Typesense is unreachable.

ACCEPTANCE: unit tests cover filter-query translation for every nautical filter; approximate listings never expose exact coordinates in any API response (assert on the JSON payload); killing Typesense locally still returns results via the fallback.

Prompt 6 — i18n, currency and units

Set up next-intl for en (default), it, fr, de, es, ru per spec §5.1.

- Per-locale routing, middleware detecting language on `/"` only (302, uncached), all other routes always prefixed, no trailing slash.
- `/src/messages/*.json` seeded with the complete copy deck from spec §11 (English), with placeholder translations for the other locales marked so they can be filled later.
- Fallback to en with a dev-time warning listing missing keys.
- LocaleSwitcher preserving path + query and mapping localised landing-page slugs correctly.
- CurrencySwitcher (EUR default) converting from `priceEur` with a daily FX snapshot and showing the original currency on hover; UnitSwitcher for `metric/imperial` stored in a cookie.
- A single hreflang helper emitting all locales + x-default on every page.

ACCEPTANCE: switching locale on a landing page keeps the equivalent page; hreflang validates; no literal user-facing string exists in any component (lint rule).

Prompt 7 — Search page

Build `/[locale]/search` per spec §10.2 and the URL contract in §5.5.

Layout: desktop 60/40 virtualised list + sticky MapLibre map; mobile list with a Map toggle to a full-screen map with a synced bottom card carousel.

Filters: every filter listed in §10.2 with live facet counts, including `boatLoa` + `draft` ("`fits my boat`": `maxBoatLoaM` `>=` `loa` AND `waterDepthAtBerthM` `>=` `draft`).

Map: water-saturated custom style, clustering above 60 markers, hover sync both ways, circles for approximate-precision listings, search-as-I-move with 350ms debounce writing `bbox` to the URL.

Sorting: featured, newest, price asc/desc, frontage desc, max boat length desc.

Empty state: compute the most restrictive filter and offer one click to relax it.

Performance: static shell, map via next/dynamic `ssr:false`, first-load JS before map < 160KB gz, skeletons with identical dimensions to results, list virtualised.

Accessibility: skip-map link, aria-live result count, labelled inputs, full keyboard operation.

ACCEPTANCE: `pnpm analyze` shows the route under budget; Playwright proves filters, `bbox`, back/forward and boat filter correctness; axe clean; Lighthouse mobile `>=` 95 on this route with 60 results.

Prompt 8 — Listing detail

Build `/[locale]/property/[slug]` per spec §10.3, SSG + ISR 600s, `generateStaticParams` over public listings only. Implement every block in the order given, including the Water credentials table and the Nautical panel with the "Fits a {n} m yacht" badge, plus the sold / under offer / expired / unlisted / sample states.

- Similar listings logic isolated in `/src/lib/similar.ts` (same `waterBodyType`, same destination or price within $\pm 35\%$, comparable frontage, exclude sold) so it can be swapped for embeddings later.
- JSON-LD `RealEstateListing` + `Offer` + `BreadcrumbList`, mapping water and nautical fields into `amenityFeature/LocationFeatureSpecification` per §14.3.
- Dynamic OG image at `/api/og/property/[slug]`.
- `viewCount` incremented server-side, session-deduplicated, never blocking render.
- Expired listings return HTTP 410 with the designed page.

ACCEPTANCE: Rich Results Test passes on a sample listing; 410 verified with curl; first-load JS < 130KB gz; gallery fully keyboard-operable with focus restored on close.

Prompt 9 — Home page

Build /[locale] per spec §10.1 and the copy deck §11.1, SSG + ISR 300s.
Hero image must be the LCP element, under 120KB AVIF, with a gradient scrim and no carousel.
Implement the "Will it fit?" strip with counts precomputed at build for 5 boat-length buckets and refined client-side on change.
ACCEPTANCE: Lighthouse mobile >= 95, LCP < 1.2s on throttled 4G, CLS < 0.02, first-load JS < 110KB gz, and the LCP element is confirmed to be the hero image in the Lighthouse trace.

Prompt 10 — Landing pages and destination hub

Build the programmatic engine per spec §5.4 and §10.4.
- /src/lib/seo/combo.ts resolving and canonicalising combinations, 301ing aliases.
- /[locale]/waterfront/[combo] rendering editorial copy + live aggregate stats + listings grid + internal links (6–10 siblings) + FAQ with FAQPage JSON-LD + lead form; published-and-non-empty gate, otherwise 404.
- /[locale]/destinations and /destinations/[slug].
- Seed the 40 landing pages listed in §14.4 with real 300–500 word editorial intros, written to answer the query directly in the first 60 words per §14.6.
ACCEPTANCE: every seeded combo returns 200 with unique title/description; an unpublished combo 404s; aggregates match a direct SQL count; no two landing pages share a title.

Prompt 11 — Multi-agency workflows: import and feeds

Implement spec §8.4, §8.5, §8.6, §8.7, §8.8.
- CSV/XLSX import with the exact column set in §8.4: upload → dry-run validation report per row with column-level reasons → confirm → background job → summary email + downloadable error CSV. Idempotent on (agency, reference). Images fetched, validated, optimised, deduplicated.
- Feed ingestion at /api/feeds/[agencySlug] with per-agency token, native JSON schema published at /schemas/listing.schema.json, plus adapters for Kyero XML and a generic mapped CSV over HTTP, and an admin field-mapping UI stored on the Agency.
- Vercel Cron every 6h polling feeds, producing a diff report (created/updated/withdrawn/errors) emailed to the agency; listings missing from two consecutive runs become withdrawn.
- Moderation queue page with side-by-side data, automated pre-checks (§8.6), duplicate matches and map pin check; approve / request changes / reject with a note.
- Expiry sweep cron: T-14 reminder email with one-click confirm; expiry moves to expired + 410.
ACCEPTANCE: importing the sample CSV twice creates no duplicates; a row with distance_to_water_m=80 is rejected with a clear reason; a Kyero sample feed imports; the moderation queue blocks a listing with 3 images; expiry sweep flips a backdated listing and its URL returns 410.

Prompt 12 — Leads, consent, routing

Build the LeadForm used on listing, landing, contact and list-with-us pages per spec §8.9 and §11.2. Fields, validation, honeypot, timing check and rate limiting (5/IP/hour) as specified; Zod schema shared client/server; consent stored with timestamp and IP.
Routing: property.agent → agency inbox → internal desk. Emails via Resend using React Email templates from §11.5, sent from our domain with reply-to the enquirer. Lead status lifecycle and a 48h unanswered reminder cron. Sample listings log the lead but never email an agency.
Add the non-blocking, deferred, granular cookie consent banner with a stored consent record.
ACCEPTANCE: a lead saves, routes and emails in staging; the banner causes zero layout shift and does not delay LCP; rate limit verified; a sample-listing lead sends no agency email.

Prompt 13 — Sample data generator

Implement spec §13.10 exactly, using the image sourcing rules in §13.2 and the description template grammar in §13.8, and including all safety rules in §13.12.
60 properties across the 12 destinations plus fjord and canal outliers, deterministic seed, idempotent upsert, --count/--destination/--wipe flags, internally consistent economics, full water and nautical variety so every filter returns results, 8–14 Unsplash images each with credit, licence, sourceUrl and the required download-tracking call, localised content from per-locale templates, 6 agencies, 12 agents, 10 landing pages, 6 journal articles.
Add npm sample:purge.
ACCEPTANCE: running the script twice produces identical data and no duplicates; every filter in the search UI returns at least 3 results; every sample listing shows the badge, emits noindex and is absent from sitemap.xml; sample:purge leaves zero sample records and zero orphan media.

Prompt 14 — SEO and AI discovery layer

Implement spec §14 in full.

- One metadata helper implementing the templates in §11.4 with fallbacks, canonical, OG/Twitter, and hreflang for all locales + x-default.
- Dynamic OG images for property and landing pages.
- Sitemap index with children split by type, 50k URLs max each, correct lastmod, excluding non-public, sample, unpublished and expired content.
- robots.txt with the AI crawler policy in §14.6, and /llms.txt + /llms-full.txt.
- JSON-LD for every template per §14.3, with Vitest shape tests.
- Lifecycle rules per §14.5: sold 90-day window then 301, expired 410, slug-change redirects, paginated search noindex beyond page 1.
- Public aggregate JSON endpoints /api/public/stats/[combo] with updatedAt.
- pnpm audit:seo reporting missing canonicals/descriptions, duplicate titles, orphan pages, sitemap URLs returning non-200, and listings with no inbound internal link.

ACCEPTANCE: audit:seo returns zero errors; Rich Results Test passes for listing, landing, article and organisation; llms.txt is served as text/plain; a changed slug 301s from the old URL.

Prompt 15 — Brochure PDF, agency dashboard, admin dashboard

A) /src/lib/pdf: localised A4 brochure per listing – cover image, logo, title, price or "Price on request", water credentials, nautical panel, key facts, 6 gallery images, static map, agency contact block, disclaimer footer. Colours and fonts read from /src/tokens. Exposed at /api/property/[slug]/brochure.pdf and as a Payload admin action.

B) Agency dashboard per spec §8.10: listings by status, items needing attention, views and leads per listing over 30 days, average response time, feed health, data export.

C) Admin dashboard: totals by status/destination/water body, data-quality panel (missing frontage or nautical data with direct edit links), top 10 by viewCount, leads per week by source, agency leaderboard, and a warning if any sample listing is indexable.

All queries via /src/lib/db, Recharts, token-aligned, read-only.

ACCEPTANCE: the PDF restyles when tokens change; dashboards load in under 1s with 1000 listings; an agency_admin sees only their own numbers.

Prompt 16 — Accessibility pass

Bring the whole site to WCAG 2.2 AA per spec §15. Audit and fix: contrast, focus visibility, keyboard operation of gallery, map, filters and dialogs, focus trapping and restoration, aria-live regions, form labelling and error summaries, heading order, target sizes, reduced motion.

Add @axe-core/playwright checks on: home, search, listing, landing, journal article, contact, list-with-us, and the admin login. Add a manual keyboard-only test script to the repo.

ACCEPTANCE: zero axe violations on all 8 routes; Lighthouse accessibility 100 on home, search and listing; a keyboard-only user can complete a search and submit an enquiry.

Prompt 17 — Performance pass

Enforce every budget in spec §12.1 (mobile, throttled 4G).

- Audit all routes for accidental dynamic rendering.
- Run @next/bundle-analyzer; eliminate or dynamic-import the largest modules; map, lightbox, charts and PDF must be ssr:false dynamic imports.
- Verify explicit sizes on every next/image, AVIF first, exactly one priority image per page.
- Confirm fonts self-hosted, subset, preloaded, max 4 files.
- Add cache-control headers, verify edge cache HIT on Vercel for listing and landing pages.
- Wire the signed /api/revalidate webhook from Payload afterChange.
- Add Lighthouse CI budget assertions to the pipeline as hard failures.

Report before/after numbers for every budget and list the top 5 remaining offenders.

ACCEPTANCE: all budgets met on home, search and listing; CI fails when a deliberate 50KB dependency is added.

Prompt 18 — Security, privacy, anti-scraping

Implement spec §16.

Security headers including a strict CSP compatible with MapLibre and Plausible; 2FA for admin and agency_admin; rate limits on all public POST routes and the search API; signed webhooks; Sentry with PII scrubbing; dependency audit in CI.

Privacy: granular consent with stored record, one-click lead anonymisation action in the admin, retention jobs per §16.5, and a documented subject-access procedure in /docs/privacy-procedures.md.

Anti-scraping: no public bulk endpoint, unguessable media URLs, Cloudflare rate rules documented, approximate coordinates never exposed, prohibition stated in the Terms.

ACCEPTANCE: securityheaders.com grade A; CSP produces no console violations on any page; a lead can be anonymised in one click and disappears from all exports.

Prompt 19 — Tests, monitoring, staging, launch

Complete spec §18, §19, §20.

- Full Playwright critical-path suite as listed in §18, including the multi-tenant isolation tests.
- Visual regression baselines on 10 viewports.
- Sentry, Plausible with the event schema in §17, external uptime monitor.
- Deploy to a password-protected Vercel staging domain; document every env var.
- Produce LAUNCH-CHECKLIST.md implementing §19 content freeze and §20 DoD, with the blocking item "no isSample listing is publicly indexable" and "SAMPLE_DATA_ENABLED=false in production".
- Produce README.md that gets a new developer running locally in under 10 minutes, and /docs/admin-guide.md explaining the backoffice for a non-technical editor and for an agency.

ACCEPTANCE: CI green end-to-end; staging reviewed against the checklist with every item ticked or explicitly deferred in DECISIONS.md.

Prompt 20 — Handover package

Produce the handover:

- /docs/admin-guide.md (editor + agency walkthrough with screenshots)
- /docs/architecture.md (diagrams of data flow, import pipeline, search sync, caching)
- /docs/seo-playbook.md (how to add a landing page, the copy rules, the internal-link rules)
- DECISIONS.md finalised
- A 10-minute screen recording script covering: publish a listing, import a CSV, approve a listing, read the dashboard, add a landing page.

ACCEPTANCE: a non-technical editor can publish a listing following the guide without assistance.

23. Appendix

23.1 All reference links

Primary: jamesedition.com/real_estate · jamesedition.com/real_estate/waterfront--all · christiesrealestate.com/lifestyles/waterfront-homes/ · sothebysrealty.com · thewaterfrontproperty.com · privateislandsonline.com · vladi-private-islands.de/en/ · buywaterfrontworldwide.com · waterfrontportfolio.com · luxuryestate.com · lionard.com

Patterns: airbnb.com · zillow.com · rightmove.co.uk · idealista.com · compass.com · yachtworld.com · boatinternational.com · linear.app · vercel.com · stripe.com

Docs: nextjs.org/docs · payloadcms.com/docs · postgis.net/documentation · typesense.org/docs · maplibre.org/maplibre-gl-js/docs · docs.maptiler.com · docs.protomaps.com · developers.cloudflare.com/images · developers.cloudflare.com/r2 · next-intl.dev · react-pdf.org · react.email · resend.com/docs · unsplash.com/documentation · pexels.com/api · schema.org/RealEstateListing · web.dev/vitals · w3.org/WAI/WCAG22/quickref · llmstxt.org

23.2 Glossary for the developer

LOA boat length overall · **beam** boat width · **draft** depth of hull below waterline · **fixed bridge** non-opening bridge that caps air draft · **riparian rights** rights of a riverfront owner · **demanio marittimo** Italian state ownership of the shoreline, granted by concession · **foreshore** land between high and low water · **LQIP** low-quality image placeholder · **ISR** incremental static regeneration · **facet** a filterable attribute with counts.

23.3 One-page summary for the client

Question	Answer
What is it?	A worldwide portal for properties with direct water access only
Why will it rank?	A taxonomy nobody else has → thousands of legitimate, data-rich landing pages, plus AI-discovery hygiene
Why will it convert?	The only search that answers "will my boat fit, and can I reach open sea?"
Why will agencies use it?	Free listings, three upload routes, a dashboard that proves the traffic
Why will it feel premium?	Editorial design system plus performance budgets stricter than every competitor
Time to live	~8 weeks with one senior developer using Cursor + Claude; ~5 weeks with two
Client effort required	Brand approval, legal entity details, DNS, staging sign-off. No photo or copy curation.

End of specification v2.0. Phases 2 and 3 — end-user accounts and alerts, CRM, semantic search, agent messaging — attach to the seams in /Lib/services, Lead.navigationPath and /lib/similar.ts with no rewrite.